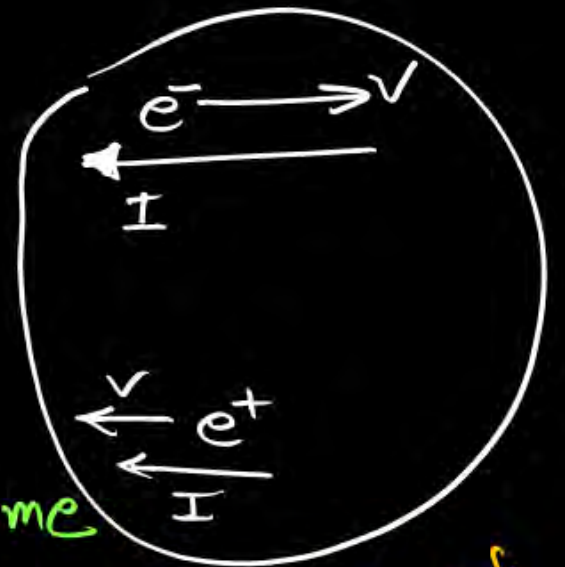


Current Electricity

Electric current — The Rate of directional flow of electric charge is c/d electric current.

in time^t

at time(t)



$$I_{Avg} = \frac{\{\Delta Q\}}{\Delta t} = \frac{(ne)_{Transferred}}{\Delta t}$$

$I_{inst} = \left(\frac{dQ}{dt}\right)$ = The rate of ^{sim} flow of charge w.r.t time

I = slope of Q/time graph
unit Amps (C-G-S → Biot)

Scalar (Tensor)

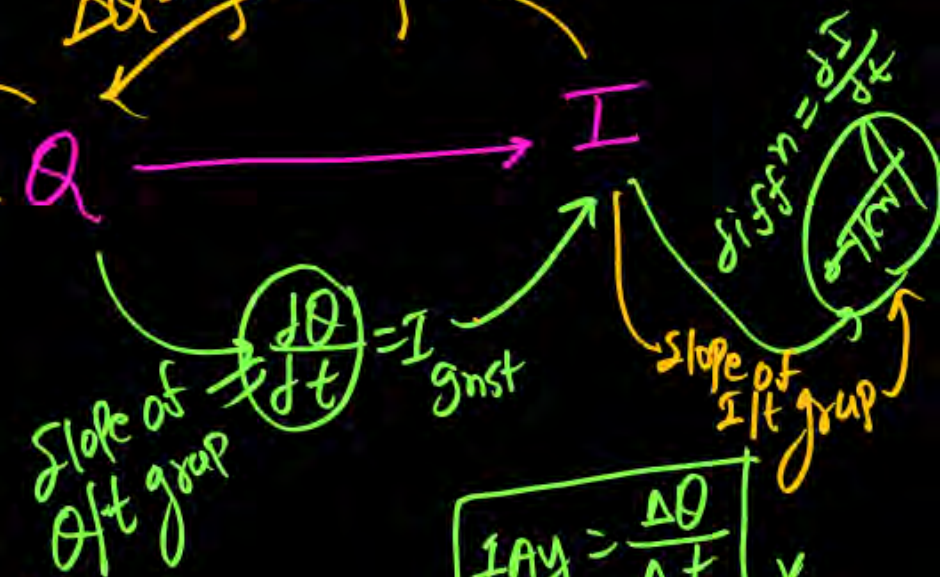
dirn along flow of the charge

does not follow Triangle law of vector addn

$$I_{Avg} = \frac{\int I_{inst} dt}{\int dt}$$

Ampere (fundamental) Physical quantity

$\Delta Q = \int \delta q = \int I dt = \text{Area of } I/t \text{ graph}$



$$I_{Avg} = \frac{\Delta Q}{\Delta t} \times$$